

PumpRupee

Pricing pump energy waste in rupees

A low-cost, clamp-on audit kit. No flow meter.

No flow meter

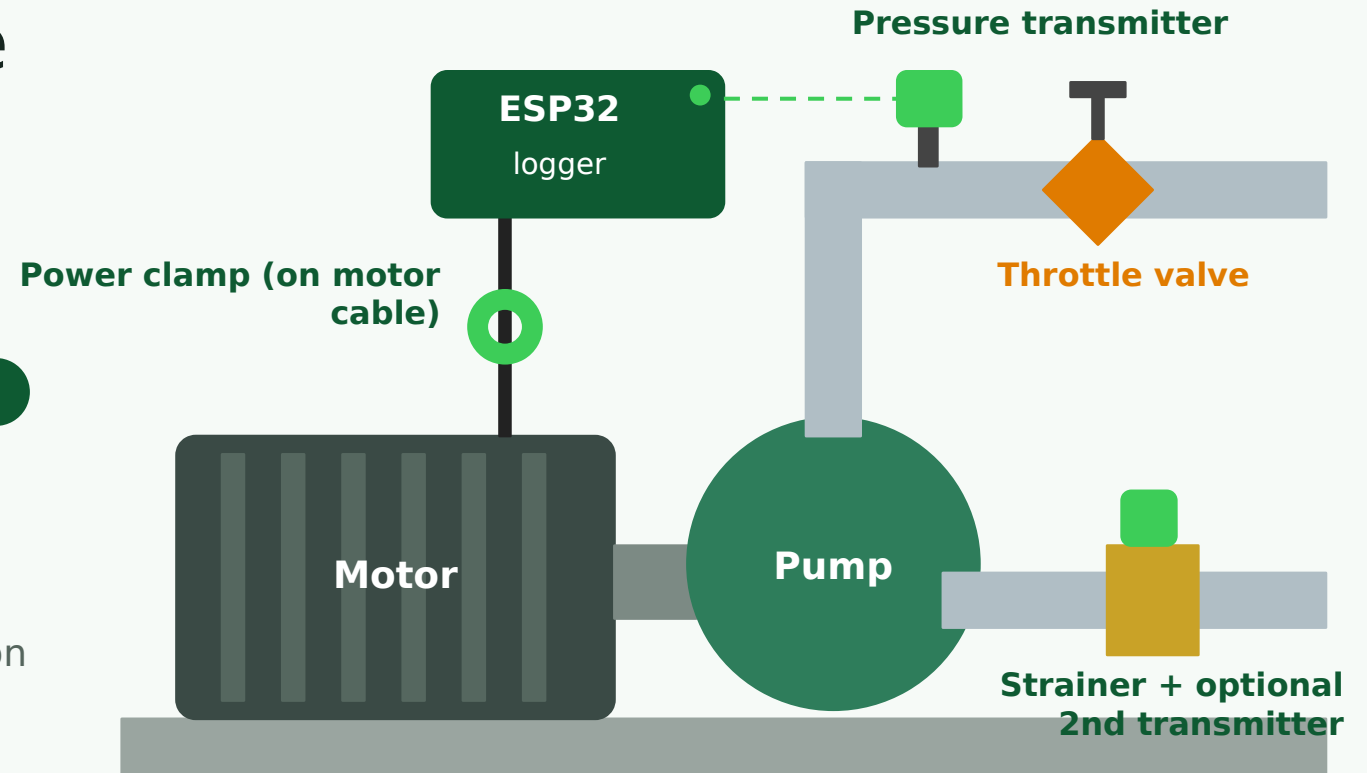
Clamp-on

₹ waste by cause

Challenge 04 - Smart Manufacturing

Yuva Yodha | Schneider Electric Energy Tech Hackathon

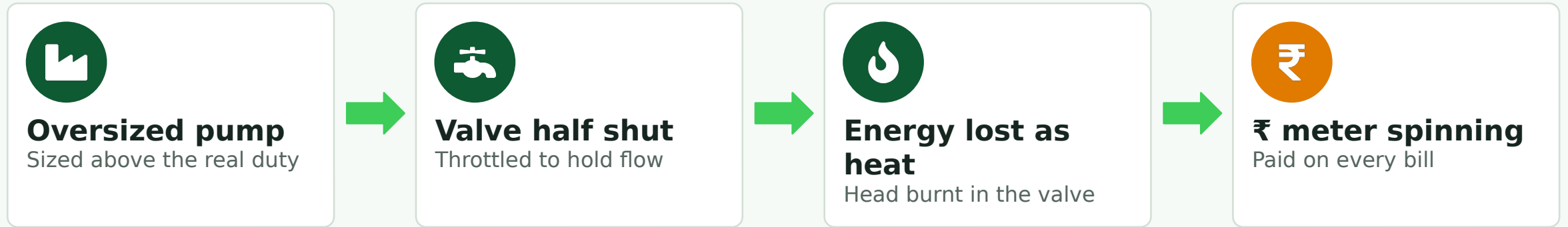
[Team name] | [Institution]



All results in this deck are simulated (Grundfos NB 65-160/157, 11 kW). Prices are indicative, not quotes.

Problem Statement

Pumps are oversized and run throttled, so energy is burnt across a valve every hour.



You pay for the pump AND for fighting the valve.

Owners can't see it because flow is hard to measure: an energy-audit practitioner reports that 30–40% of ultrasonic flow readings go wrong.

Existing approaches, and where each fits

Flow-meter retrofit

Fits: a direct flow reading.
Falls short: meter cost and installation; ultrasonic readings misread 30–40% (practitioner report).

Full ESCO audit

Fits: expert judgement and a trusted baseline.
Falls short: a point-in-time exercise and costly, not continuous.

Vendor platforms

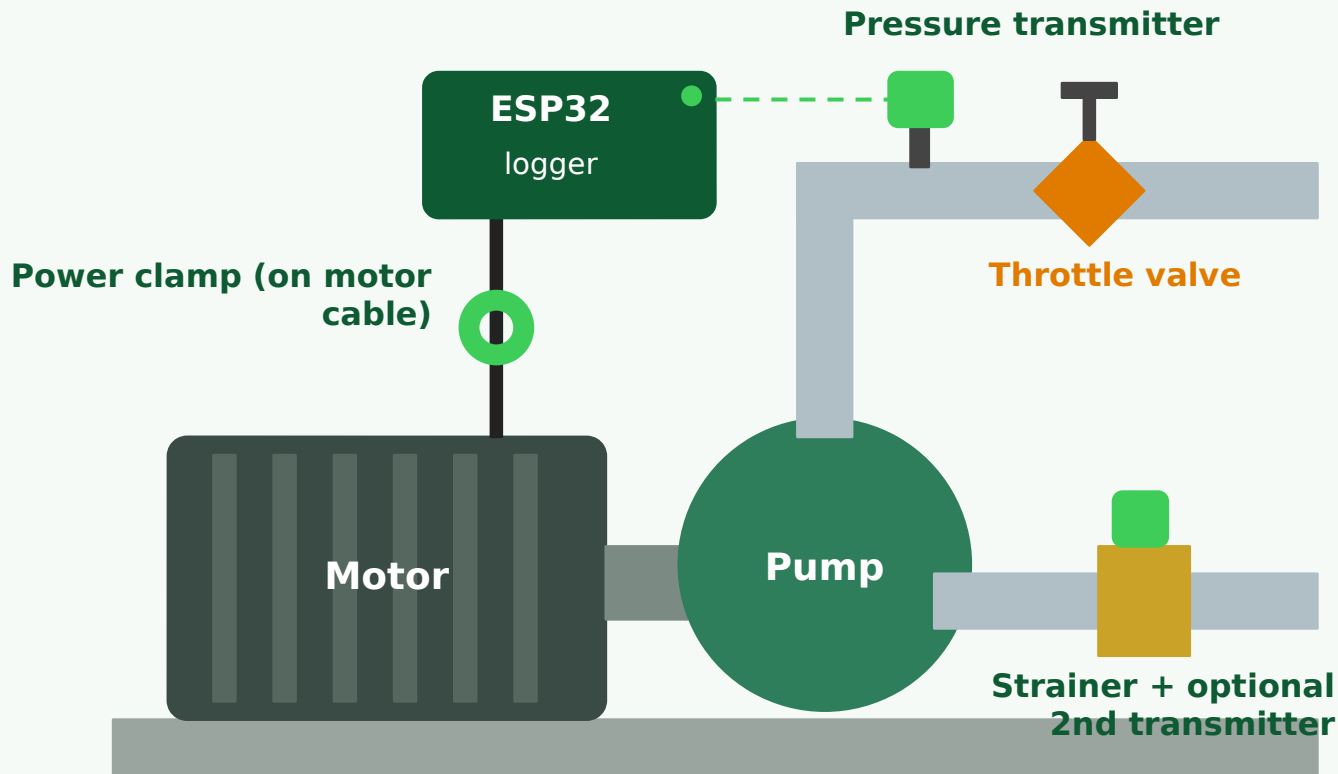
Fits: deep integration with the vendor's own equipment.
Falls short: often vendor-locked or subscription-based.

PumpRupree

Fits: ₹ waste by cause at ₹6,100–13,300 per kit; no flow meter.
Limit: audits only, and results are simulated so far.

Proposed Solution

A clamp-on kit turns power + pressure readings into “₹ wasted per year, and why”.



₹ waste per year

The pump's energy waste priced in rupees



Waste by cause

Valve, impeller wear, clogged strainer



Ranked fixes + setpoint

Cheapest fix first; lowest safe VFD setpoint



Verified saving

Re-measure after the fix

Audits and recommends. Does not control the pump.

Alignment to Challenge 04 - Smart Manufacturing

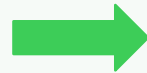
Asset-level energy visibility, priced in rupees, at MSME entry cost.

Challenge goal

PumpRupee feature



Energy visibility at the asset level



SEC (kWh/m³) and ₹/m³ for each pump, plus the waste split by cause



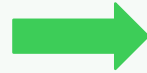
Data-driven maintenance



CUSUM wear-drift alarm (wear caught at about 2–4%); strainer clog made visible with one extra pressure sensor



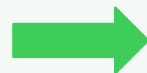
Measurable kWh and CO₂ reduction



Before/after re-measurement; CO₂ at 0.710 kg/kWh (CEA); simulated 6.1–11.3 t CO₂/yr per 11 kW pump



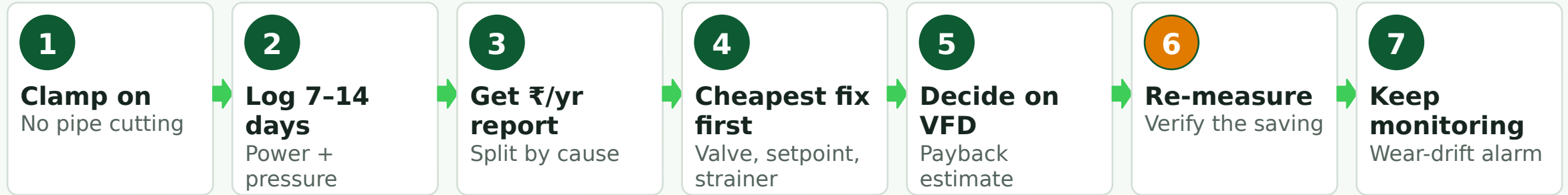
Low-cost entry for MSMEs



Clamp-on kit from readily available parts, about ₹6,100–13,300; no flow meter

Key Features & User Journey

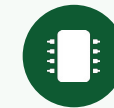
From clamp-on to a verified saving, in seven steps.



Owner report: Pump P-01

EXAMPLE LAYOUT | SIMULATED VALUES

Waste by cause	Valve 83-93% of the waste; wear and clog share the rest
Top fixes, in order	1. Strainer, grease, alignment (cheap) 2. Speed control or impeller trim (largest saving)
Recommended setpoint	24.2 m (2.37 bar), constant pressure
VFD saving	19.3% = ₹69,292/yr, 6.1 t CO ₂ /yr
VFD payback	8.7-18.9 months. Example verdict: worth buying



No flow meter

Flow inferred from pump curve, power and head



Clamp-on install

Off-the-shelf parts, no pipe cutting



VFD verdict before you buy

Payback for YOUR pump



Before/after verification

Re-measure after the fix

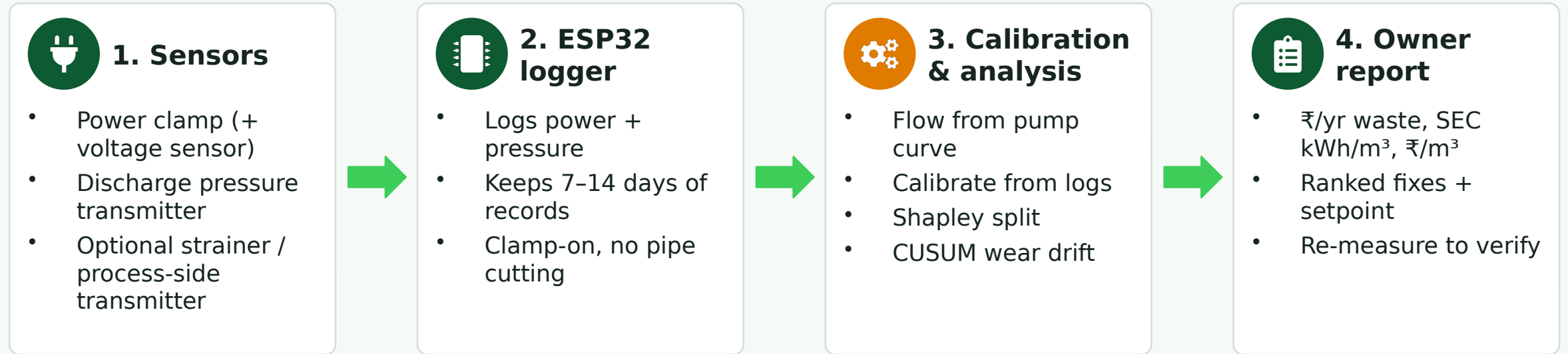


Wear-drift alarm

Catches slow impeller wear

Technical Approach: Architecture & Method

Sensors → ESP32 logger → calibration & analysis → owner report.



Method in plain words

Flow without a meter

Pump curve + logged power + head give an estimate of flow.

Calibration

Learn from 7-14 days of logs. Optional anchors: shut-off head, closed-valve power.

Shapley split

A fair way to share the blame among valve, wear and clog.

CUSUM drift

An alarm that notices slow drift, such as impeller wear.

SEC

Units of electricity per m³ pumped (kWh/m³).

Flow is inferred, never measured. A bad flow reading ($\pm 30\%$) is worse than no calibration, so PumpRupe does not depend on one.

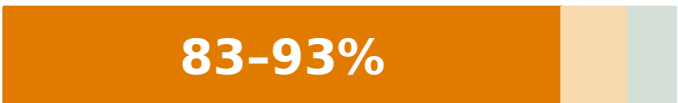
Technical Approach: Simulated Validation

SIMULATED - Grundfos NB 65-160/157, 11 kW, IE3

Assumed: 10 m static lift | 5% head margin | 16 h/day | 300 days/yr | ₹8/kWh | VFD efficiency 0.97 | 0.710 kg CO₂/kWh (CEA)

Where the waste goes

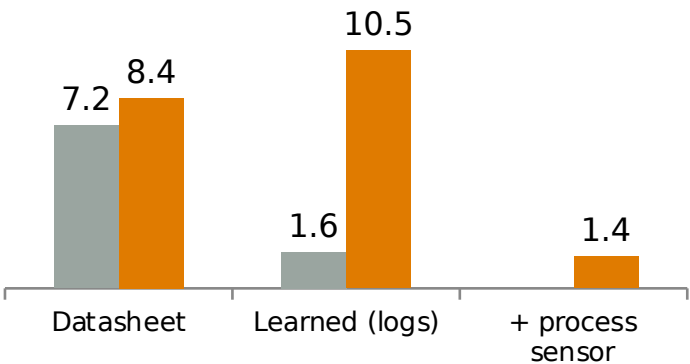
Valve (throttling) share of waste



Rest: wear, clog, other. Under stress tests the valve share is 72-87%. Measured against a speed-matched, healthy, clean ideal.

Clog blind spot
A clogged strainer is invisible with power + pressure alone. One extra pressure sensor recovers about ₹6,000-6,600/yr of misattributed waste.

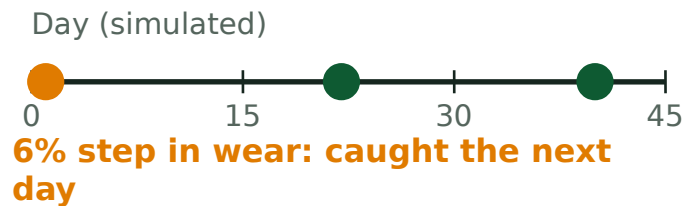
Forecast accuracy



■ Friendly test (1% noise)
■ Stress test (3% noise, curve errors)

Mean error, percentage points.
The system curve, not the pump curve, is the main error source. A $\pm 30\%$ flow reading is worse than none (16.8 vs 8.3).
About ± 3 points only with a measured system curve; else about ± 10 .

Wear-drift detection



Fast wear: alarm ~day 22 at $\approx 3.7\%$ wear

Slow wear (0→10% in 180 days): alarm ~day 40 at $\approx 2.2\%$ wear

Wear caught at about 2-4%
1 false alarm in 3×365 simulated days

Innovation: System-Level Packaging

We do not claim pump monitoring, sensorless flow or VFD savings are new.

Existing technology (we build on it)

- **KSB PumpMeter** pump monitoring
- **US DOE PSAT** pump energy assessment tool
- **Samotics** motor-current pump efficiency
- **Schneider Altivar Process** sensorless flow in the drive
- **Indian IoT vendors, ESCOs** monitoring and audits



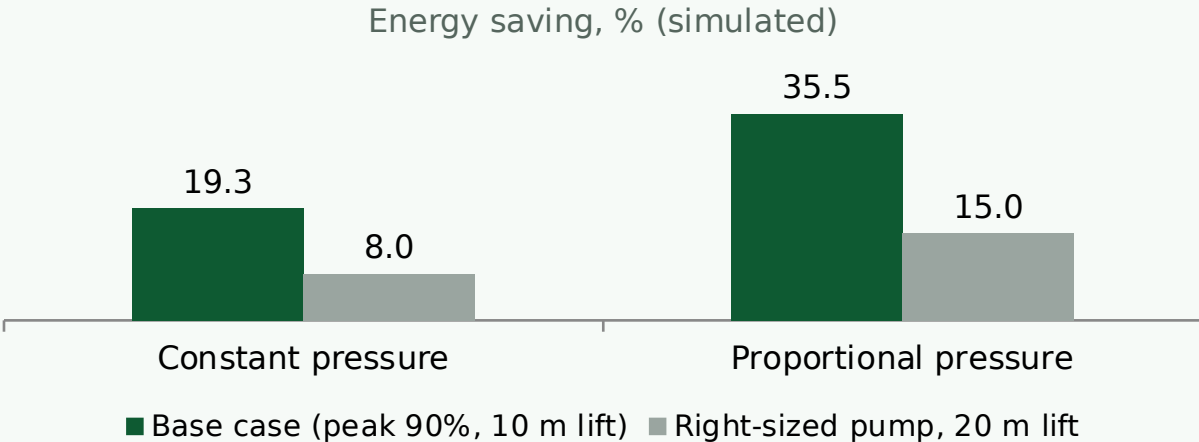
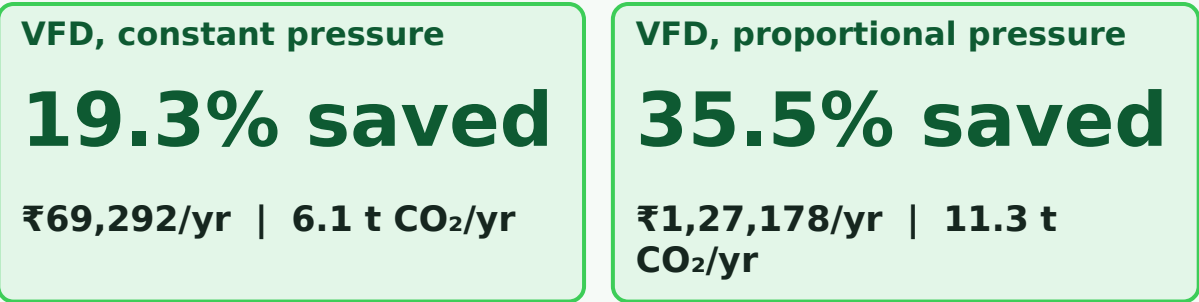
Our integration

- ✓ ₹/year waste split by cause, not just an efficiency number
- ✓ Prices the extra sensor that would reveal what the basic kit cannot see
- ✓ Low-cost shared audit kit with no flow meter
- ✓ Before/after verification built in
- ✓ A VFD payback verdict for the owner's own pump

We found no low-cost shared kit that combines all four. (A search result, not proof that none exists.)

Expected Impact: Savings & Cost

SIMULATED - per 11 kW pump, peak demand 90% of rated flow



Range across cases: 8-33% (constant), 15-45% (proportional).
Heavily oversized pumps reach 54-60% in the model: an upper bound only.
An energy-audit practitioner reports 5-40% in practice.

Who benefits: MSME plant owners, maintenance teams, energy auditors. **Scale-up (illustration, not a projection):** N pumps × per-pump saving. For N = 10 on the base case: ₹6,92,920/yr and 61 t CO₂/yr (constant pressure).

Cost & payback (indicative prices)

Basic single-phase	₹6,100
3-phase + 2nd transmitter	₹10,200
3-phase + process-side transmitter	₹13,300
With ₹12,000 industrial transmitters	₹40,100

VFD for 11 kW: ₹29,000-94,000 + ₹15,000 install (assumed)

VFD payback (simulated)

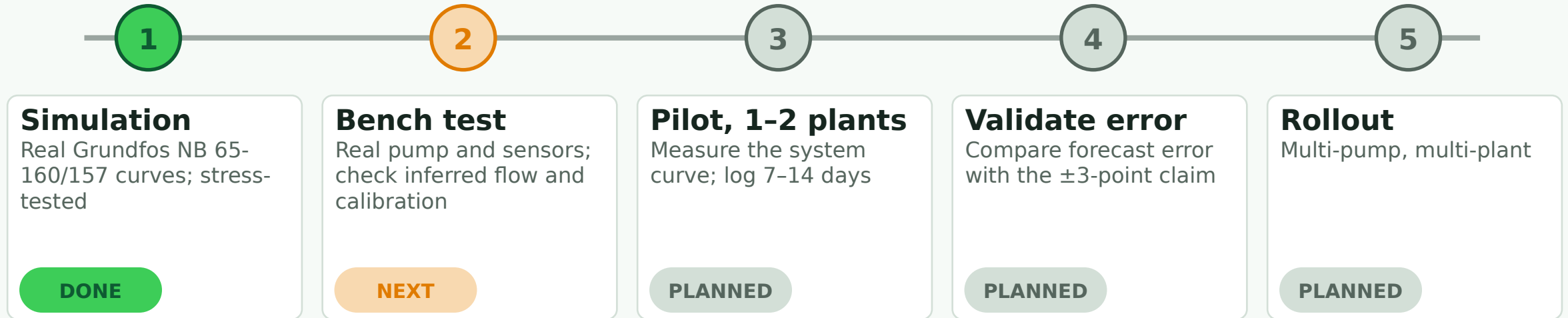


The audit tells you which case you are in BEFORE you spend on a VFD.

Audit kit cost is shared across pumps and is not included in the payback.

Implementation Roadmap

From a simulated result to a field-validated audit kit.



What we validate next (honest limits)

- Simulation only: no lab or field data yet, and no real quotes
- About ± 3 points only with a measured system curve; otherwise about ± 10
- The system-curve fit uses the same form as the simulated plant, so it is optimistic
- Single pump, steady state, hourly. Next: parallel pumps, closed loops, pumps already on drives, minimum-flow and NPSH checks, control dynamics

Team Introduction

[Team name] | [Institution]



[Name]

[Role]

[Skill: one line relevant to
pumps, sensors, data or
energy]



[Name]

[Role]

[Skill: one line relevant to
pumps, sensors, data or
energy]



[Name]

[Role]

[Skill: one line relevant to
pumps, sensors, data or
energy]



[Name]

[Role]

[Skill: one line relevant to
pumps, sensors, data or
energy]

Replace the photo circles and [placeholders]; delete unused cards (1-4 members).